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Lesson Notes Part II

MODULE 1 | LESSON 1 | PART II

SAMPLE MOMENTS AND PORTFOLIO PERFORMANCE - PART II

Reading Time	60 minutes
Prior Knowledge	Basic statistics and a working knowledge of basic portfolio theory
Keywords	Return and variance of a portfolio, Beta of a portfolio and relevant types of risk, Performance appraisal (Sharpe and Treynor ratios, Sortino ratio, information and appraisal ratio, skewness, and kurtosis)

This lesson introduces financial portfolios and discusses how to compute certain moments like mean and standard deviation as well as a measure of systematic risk, like the beta. Then, after a discussion of risk, we look at how to assess the performance of portfolios in regards to particular risks. At this stage, the traditional indices used for performance appraisal are introduced, like the Sharpe and Treynor ratios. Furthermore, we discuss how volatility is bad only when it is on the downside and not on the upside, and we also consider other measures of appraisal. Finally, we discuss the implications of skewness and kurtosis and what impact they could have on performance if assessed without regard to these statistics.

5. Skewness: The Sortino Ratio

Traditional investing theory assumes that investment returns form a symmetrical, bell-shaped curve, with most of the returns in the middle, and large gains and losses falling at the “tails” of the distribution. In reality, return distributions are often tilted a bit—and sometimes more than a bit to the left. This increases the likelihood of “disappointing returns” that are lower than expectations. If one assumes a normal distribution, this assumption fosters a sense of confidence that the portfolio's returns have a low probability of falling more than three standard deviations from the mean, less than 0.15% as a matter of fact. Also, under the assumption of normal returns, the Sharpe ratio provides a reliable appraisal measure to assess the performance of a portfolio. However, the empirical evidence shows that several classes of assets exhibit negative skewness, and when this negative skewness is not negligible, this can become a significant problem.

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portfolios when skewness of returns is taken into consideration.

Regarding the first problem, we notice that the Sharpe ratio does not differentiate between good and bad volatility and does not take skewness into consideration. Most, if not all, investors would not object when the returns are volatile above a certain threshold on the upside. Then, again, most if not all would feel very uncomfortable when volatility hits their portfolios on the downside. Essentially, one drawback of the Sharpe ratio is that the portfolio manager is penalized for all volatility, good and bad, and does not differentiate between the "good" and the "bad" tails.

The Sortino ratio considers the standard deviation on the downside, i.e.

$$\mathrm{SoR}_p = \frac{\bar{r}_p - r_{MAR}}{\sigma_{downside}}.$$

In this case, a target rate of return, MAR (minimum accepted return), is identified and $\sigma_{downside}$ refers to the target semi-standard deviation or the volatility of only those returns that are below the MAR. The Sortino ratio is more appropriate for investments that do not exhibit returns with symmetrical distributions. In this case, using Sharpe ratio, positively and negatively skewed returns would both produce a lower Sharpe ratio. However, only investments with negatively skewed return distributions would result in a lower Sortino ratio. While the Sortino ratio is convincing at dealing with the problem of positive and skewed volatility, the target return is somewhat arbitrary, unlike the risk-free rate, and different investors could define it differently, ending up with different calculated Sortino ratios.

There is also an M^2 version of the Sortino ratio.:

$$M^2_{p,Sortino} = r_p + \mathrm{SoR}(\sigma_{\mathrm{Dbench}} - \sigma_D)$$

where r_p is the annualized portfolio return, σ_{Dbench} is the benchmark annualized downside risk and the σ_D the portfolio annualized downside risk.

Regarding the second difficulty that we raised, that is how to take skewness into consideration when determining the optimal portfolio: One has to introduce the notion of co-skewness. This will be done in the next module. Here, we introduce the definition of skewness, which, unlike how variance or standard deviation involves the second moments of distributions, involves the third moment of distributions.

Skewness is a measure of the asymmetry of a distribution. Negative skewness occurs when the values to the left of the mean are fewer but further from it than the values to the right. Positive skewness occurs if the opposite is true. Skewness (Sk) for a collection of returns $r_1, r_2, \ldots, r_N$ is calculated as

$$\widehat{\mathrm{Sk}} \equiv \hat{\gamma} = \frac{\sum_{i=1}^{N}(r_i - \bar{r})^3}{(N-1)s^3}$$

For samples of points that are not very large, the following modified formula has better statistical properties:

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In both cases, $\bar{r}$ is the sample mean and s the sample volatility of the returns. For large values of n, the two estimates are almost identical.

5.1. Example: The Skew-Normal Distribution

We say that the random variable X has the skew-normal distribution, $X \sim SN(\mu, \sigma, \alpha)$ where μ is the location parameter, $\sigma > 0$ is the scale parameter, and α is the shape parameter, if its probability density function has the following form:

$$f(x) = \frac{2}{\sigma}\phi\left(\frac{x-\mu}{\sigma}\right)\Phi\left(\alpha\left(\frac{x-\mu}{\sigma}\right)\right)$$

with $\phi(\cdot)$ and $\Phi(\cdot)$ being the probability density function (PDF) and the cumulative distribution function (CDF) of a standard normal random variable, respectively.

Note that, unlike in the case of a $N(\mu, \sigma^2)$ random variable, μ and σ are not the mean and the standard deviation of the distribution, except when $\alpha = 0$. As a matter of fact, if $X \sim SN(\mu, \sigma, \alpha)$, then

$$E[X] = \mu + \sigma\delta\sqrt{\frac{2}{\pi}}$$

and

$$\text{variance}[X] = \sigma^2 \cdot \left(1 - \frac{2\delta^2}{\pi}\right)$$

with

$$\delta = \frac{\alpha}{\sqrt{1+\alpha^2}}.$$

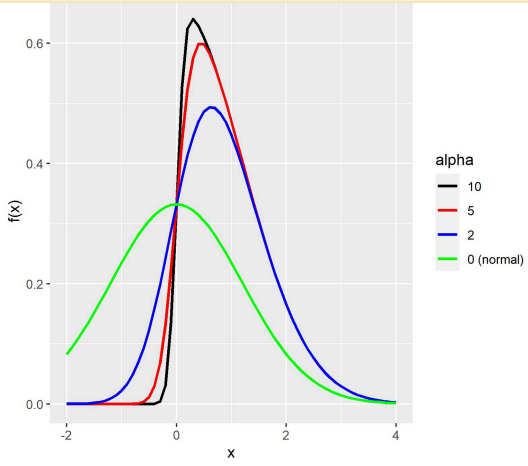
When $\alpha < 0$, the distribution is negatively skewed, and it is positively skewed if $\alpha > 0$. The formula for the skewness is somewhat complicated, but it is available:

or, in terms of just α

$$\gamma_1 = \frac{4-\pi}{2}\left(\frac{\sqrt{2}\alpha}{\sqrt{\pi(1+\alpha^2) - 2\alpha^2}}\right)^3.$$

Furthermore, the rate of decay to 0 of the probability density function as $|x| \to \infty$ is either the same as that of the normal density or even faster, depending on whether α and x have equal or opposite signs. This means that the skew-normal retains normal-like tails in the direction of the skew. A consequence of this fact is that a skew-normal is useful when the distribution of returns has no more outliers than the normal distribution. For this very reason, the skew-normal distribution is unsuitable in applications where the observed data exhibits heavier tails than the normal distribution. This means that when dealing with financial portfolios, the skew-normal family can address problems of skewness but not those of kurtosis (which we will be looking at below).

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One can observe how the tails of skew-normal distributions behave using the following animation (change the parameters to see the changes in real time):

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The ability to use a skew-normal distribution to fit the returns of several stocks is weak (change the parameters to see the changes in real time):

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Maximum likelihood estimates for μ, σ and α can be computed numerically, although no closed-form expression is available. For an independent sample $x_1, x_2, \ldots, x_n$ the log-likelihood function is given by:

$$l(\mu, \sigma, \alpha; x_1, \ldots, x_n) = \text{ constant } - n \ln(\sigma) - \sum_{i=1}^{n} \frac{(x_i - \mu)^2}{2\sigma^2} + \sum_{i=1}^{n} \xi_o\left(\frac{\alpha(x_i - \mu)}{\sigma}\right),$$

with

$$\xi_0(x) = \ln(2 \cdot \Phi(x)).$$

To complete this section, we'll look at one more aspect. When we consider a sequence of returns for an investment or a portfolio, we don't work with the population of all possible returns, but we have a sample of them available. If we compute the skewness using the formula given above, the result could change with the particular sample used. In other words, the sample skewness is a random variable. The practical implication is that we should test whether it is zero or not. This can be done easily:

$$H_0 : \gamma = 0 \quad \text{vs.} \quad H_1 : \gamma \neq 0.$$

Using $\hat{\gamma}$ obtained from the sample, if the number of observations N is large and under the assumption of normality, the distribution of the sample skewness is itself normal with mean 0 and standard error $\text{S.E.}(\hat{\gamma}) = \sqrt{6/N}$. This means that we can test whether the skewness is zero or not using the following test-statistics:

$$z = \frac{\hat{\gamma} - 0}{\sqrt{6/N}}$$

which has a standard normal distribution.

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(Students may like to peruse the following paper by Rui Albuquerque: "Skewness in Stock Returns: Reconciling the Evidence on Firm versus Aggregate Returns")

6. Information and Appraisal Ratio

The **information ratio** (IR) is used to measure the performance of a portfolio against a benchmark while accounting for the differences in overall risk. It is defined as:

$$IR_p = \frac{\bar{r}_p - \bar{r}_b}{\sigma(r_p - r_b)},$$

with $\bar{r}_p, \bar{r}_b$ representing the mean return for the portfolio and the portfolio benchmark, respectively. In this case, the denominator $\sigma(r_p - r_b)$ can be interpreted as the **tracking risk** or the volatility of the portfolio returns vs. that of its benchmark. This characteristic makes the information ratio particularly suitable for comparing the performance of portfolio managers who were assigned pre-defined benchmarks.

The appraisal ratio is a form of signal-to-noise ratio used to measure a fund or portfolio's performance. It is calculated by taking the alpha of the investment of the portfolio and dividing it by its non-systematic risk. The alpha represents excess return, calculated as the return earned by the portfolio over the return implied by the CAPM model. Its formula is $AR_p = \frac{\alpha}{\sigma_u}$.

Now, the alpha is measured by comparing the return in excess of that required by the CAPM, i.e.,

$$\alpha = r_p - (r_{RF} + \beta_p(r_m - r_{RF}))$$

and
σ_u is as it was derived in part 3 of this lesson, i.e.: $\sigma_u = \sigma^2_{r_p}(1 - corr^2(r_p, r_m))$.

7. Kurtosis and the Problem of Heavy Tails of Distribution

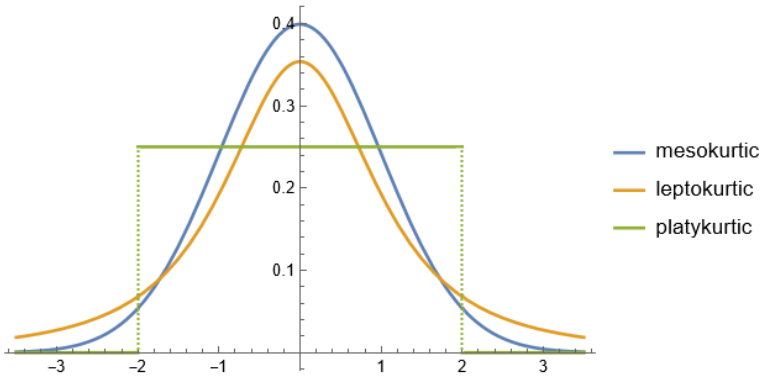
In statistics, the term kurtosis is used to describe the distribution of the dataset, and that makes it possible to measure to what extent the data points of a particular distribution differ from the data of a normal distribution. We are aware that for a normal distribution, less than 0.15% of its values fall above or below three standard deviations from the mean. However, if a distribution has heavier tails than the normal, it could have more extreme values. Kurtosis can also be combined with skewness.

In finance, kurtosis is used to measure the amount of financial risk associated with any instrument or transaction. The more kurtosis, the higher the financial risk associated with the concerned dataset. We could say that, while skewness is a measure of symmetry in distribution, kurtosis is the measure of heaviness of the tails of distributions. We distinguish distribution functions according to which of the three types of kurtosis they exhibit:

- mesokurtic distribution if the distribution has a kurtosis statistic similar to that of the normal distribution;
- leptokurtic distributions if the tails of the distributions are longer than those for the normal distributions, i.e., the population has more outliers than the normal distribution; and

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Figure 2: Different Types of Kurtosis



The formula for the kurtosis (K) of a random variable X is given by:

$$\mathrm{K}(X) = \left[\left(\frac{X-\mu}{\sigma}\right)^4\right] = \frac{\mu_4}{\sigma^4}$$

which tells us that kurtosis is the fourth standardized moment of the distribution of X with μ_4 the fourth central moment and σ the standard deviation.

The theoretical formula used in the definition above is good, but the required moments μ_4 and σ are seldom known, and one needs to use the sample kurtosis, which for a sample of size N, $X_1, X_2, \ldots, X_n$ is defined as

$$\widehat{\mathrm{K}}(X) = N \cdot \frac{\sum_{i=1}^{N}(X_i - \bar{X})^4}{(\sum_{i=1}^{N}(X_i - \bar{X})^2)^2}.$$

There is another formula with better properties for estimating kurtosis:

$$\widetilde{\mathrm{K}}(X) = \frac{N(N+1)(N-1)}{((N-2)(N-3)} \cdot \frac{\sum_{i=1}^{N}(X_i - \bar{X})^4}{(\sum_{i=1}^{N}(X_i - \bar{X})^2)^2}.$$

Again, for large values of N, the two formulae yield very similar estimates.

The normal distribution has a kurtosis $= 3$. Then, we could say that a distribution is platykurtic if its kurtosis is < 3 and leptokurtic if its kurtosis is > 3, instead. Sometimes, out of convenience, we use the **excess kurtosis**, which is simply kurtosis -3. In this way, we can talk of mesokurtic distributions if their excess kurtosis is near 0, platykurtic when the distribution has negative excess kurtosis, and leptokurtic when the distribution has positive excess kurtosis.
The uniform distribution has an excess kurtosis of -1.2, and the logistic distribution has an excess kurtosis of 1.2 instead.
The t-distribution with ν degrees of freedom has an excess kurtosis $\frac{6}{\nu-4}$ for $\nu > 4$, infinite for $\nu = 3, 4$, and undefined for $\nu = 1, 2$.

The binomial distribution with parameters n and p is found to have an excess kurtosis equal to $\frac{1-6p(1-p)}{np(1-p)}$, which can go from positive to negative depending on the value of the parameter p.
In finance, the main problem associated with the "heaviness" of tails is that using the normal

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7.1. Example: Student's t-distribution

A probability often used to represent situations where the distribution of returns has heavier tails than a normal one is Student's t-distibution. The functional form of its PDF is

$$f(x) = \frac{\Gamma((\nu+1)/2)}{\sqrt{\nu\pi}\Gamma(\nu/2)}\left(1+\frac{x^2}{\nu}\right)^{-(\nu+1)/2}.$$

where $\nu > 2$ is a parameter that represents the degrees of freedom. We write $t_{(\nu)}$ or $t(\nu)$ to denote this distribution. It has mean 0 and its variance is equal to $\frac{\nu}{\nu-2}$.

a$t_{(\nu)}$ distribution is symmetrical about the mean (in our case 0 although there are versions of Student's t also for non-zero means, non-central t-distribution) like the normal distributions, and therefore it has zero skewness. However, it has heavier tails than the normal distribution has.

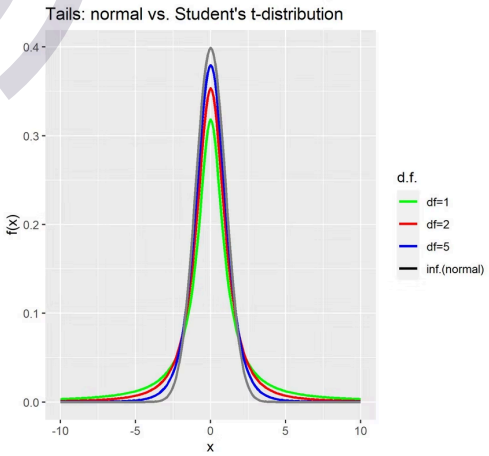
Another important feature is that as the number of degrees of freedom increases towards ∞, the distribution becomes closer to a normal distribution. In other words, the standard normal distribution can be considered the limit of a Student's t-distribution as the degrees of freedom increase to infinity. For most practical purposes, the approximation is already strong even for $\nu = 100$ and perhaps even just $\nu = 60$.

Like for the normal, if it is desired to retain certain features of the Student's t-distribution in terms of heaviness of the tails, but we want to introduce some skewness in our applications, there is a version of this distribution called the skewed t-distribution. It is actually easy to define if practitioners need it for their work:

$$f(x, \nu, \alpha) = 2 \times f(x, \nu) \times T(\alpha \cdot x, \nu),$$

with ν the degrees of freedom, $T(\cdot, \nu)$ the distribution function of a Student's t-distribution with ν degrees of freedom, and α a shape parameter that introduces a slant in the distribution.

Figure 3: Normal vs. Student's t-distribution Tails



As we did in the case of the sample skewness, we are interested in testing whether the distribution of, say, a specific sequence of returns exhibits fat tails or kurtosis. We would like to perform the test:

$$H_0 : k = 0 \quad \text{vs.} \quad H_1 : k \neq 0,$$

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$$z = \frac{k - 0}{\sqrt{24/N}}$$

which has a standard normal distribution.

It is possible to test for normality by combining the test for skewness and kurtosis into a single test. In this case, the test is known as the Jarque-Bera test:

$$\text{JB } = N\left(\frac{\hat{\gamma}^2}{6} + \frac{(\hat{k} - 3)^2}{24}\right)$$

and which has a chi-square distribution with 2 degrees of freedom.

The skewness-kurtosis ratio is the division of skewness by kurtosis.

$$\text{Skewness-Kurtosis ratio} = \frac{\text{Skewness}}{\text{Kurtosis}}$$

It is used in conjunction with the Sharpe ratio to rank portfolios. The higher the rate, the better. In fact, since kurtosis is always positive, the ratio being bigger means that there is more "good tail" in our distribution of returns.

There exists also an adjusted Sharpe ratio (introduced by Pezier and White in 2006) to adjust for skewness and kurtosis by incorporating a penalty factor for negative skewness and excess kurtosis:

$$\text{adjusted-Sharpe ratio} = SR \cdot \left[1 + \frac{Sk}{6} \times SR - \frac{K - 3}{24} \cdot SR^2\right]$$

where SR is the Sharpe ratio (using annualized data), Sk the skewness, and K the kurtosis of returns.

The performance measures presented in this lesson are just a sample of many, although they are among the best known and most used. The lesson one should take away is that there are many ways to assess portfolio performance, and each has its strengths and weaknesses. But when used in conjunction, and not blindly, they can often provide a more complete picture.

8. Conclusion

We have gained an appreciation for sample moments and betas and how they can help to appraise the performance of a portfolio. We have also seen that skewness and kurtosis of returns create problems; in particular, the behavior of the tails of the probability distribution of returns can make an analysis based on bell-shaped distributions less valid or even misleading. In the next lesson, we will look at how to measure the risk on the tails of return distributions when we look at Value at Risk (VaR).

References

- Albuquerque, Rui. "Skewness in Stock Returns: Reconciling the Evidence on Firm Versus Aggregate Returns ." *Review of Financial Studies*, vol. 25, 2011, pp. 1–56.

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